WASP-1b

R-1459

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_131028` · created 2026-08-01T10:33:52+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456593.7287 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.023
Transit depth (Rp/Rs)^2	0.82 +/- 0.41 [%]
Orbital Inclination (inc)	89.71 +/- 1.28
Airmass coefficient 1 (a1)	1.0715 +/- 0.0084
Airmass coefficient 2 (a2)	-0.059 +/- 0.019
Scatter in the residuals of the lightcurve fit is	0.78 %
Best Comparison Star	None
Optimal Aperture	3.57
Optimal Annulus	6.23
Transit Duration (day)	0.1516 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.09 ± 0.023 vs 0.1036 (deviation 0.49σ)
Duration fit vs published	0.1516 d vs 0.1567 d (deviation 1.16σ)
Mid-time O–C	-13.3 min
Depth SNR	3.2
Residual scatter	0.78 %

Light curve

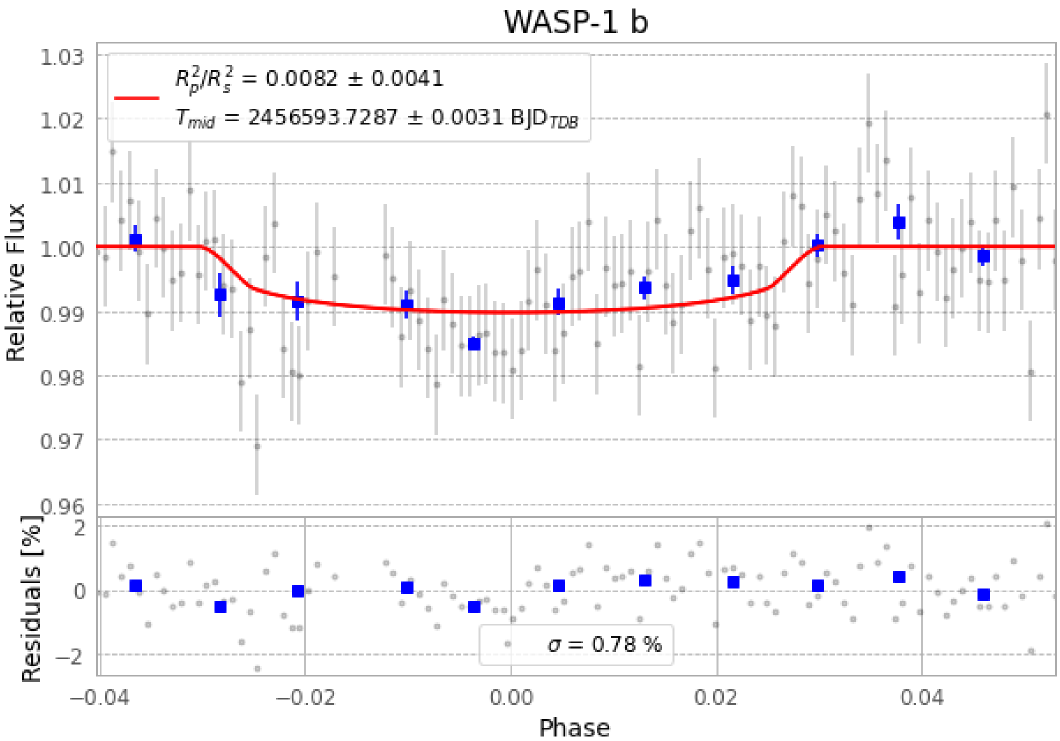

Figure 1 — FinalLightCurve_WASP-1 b_2013-10-27.png (SHA-256 b179956d1920bbfb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 259], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 a671e1d2532c7b26…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

114 FITS frames (75 MB), `WASP-1131028025412.FITS` → `WASP-1131028083312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-131028.log (0b19edf3bc63…), FinalLightCurve_WASP-1 b_2013-10-27.png (b179956d1920…), FinalLightCurve_WASP-1 b_2013-10-27.csv (32683cdfd440…)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)